



Neural Networks, CPU & GPU – Notes for Learners

A beginner-friendly guide to understanding how neural networks work, what deep learning is, and why GPUs are the engines powering modern AI.



1. What is a Neural Network?

A Neural Network is a machine learning model inspired by the human brain. It learns patterns from data, recognizes relationships, and makes predictions — without being explicitly programmed with rules.

Traditional Programming

Input + Rules → Output

Humans write all the rules manually.

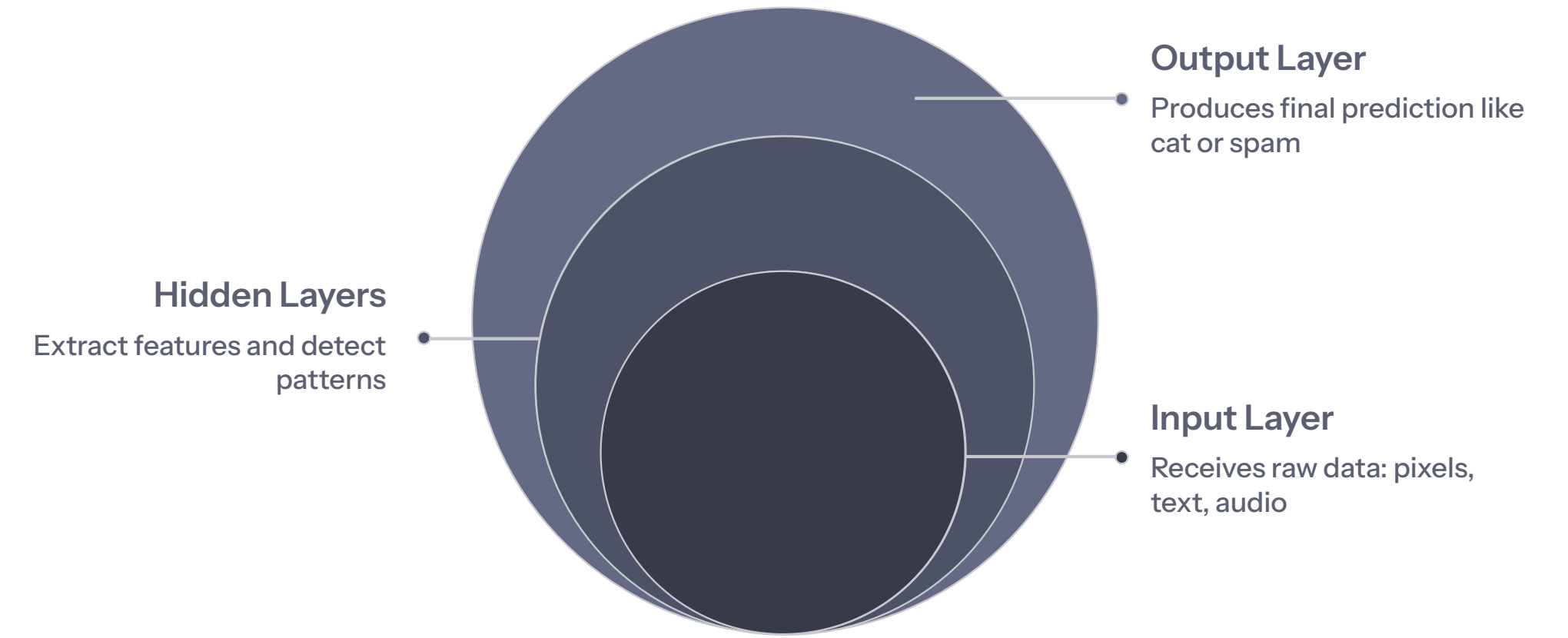
Neural Network Approach

Input + Data → Learning → Prediction

The model discovers patterns automatically from examples.



2. Structure of a Neural Network

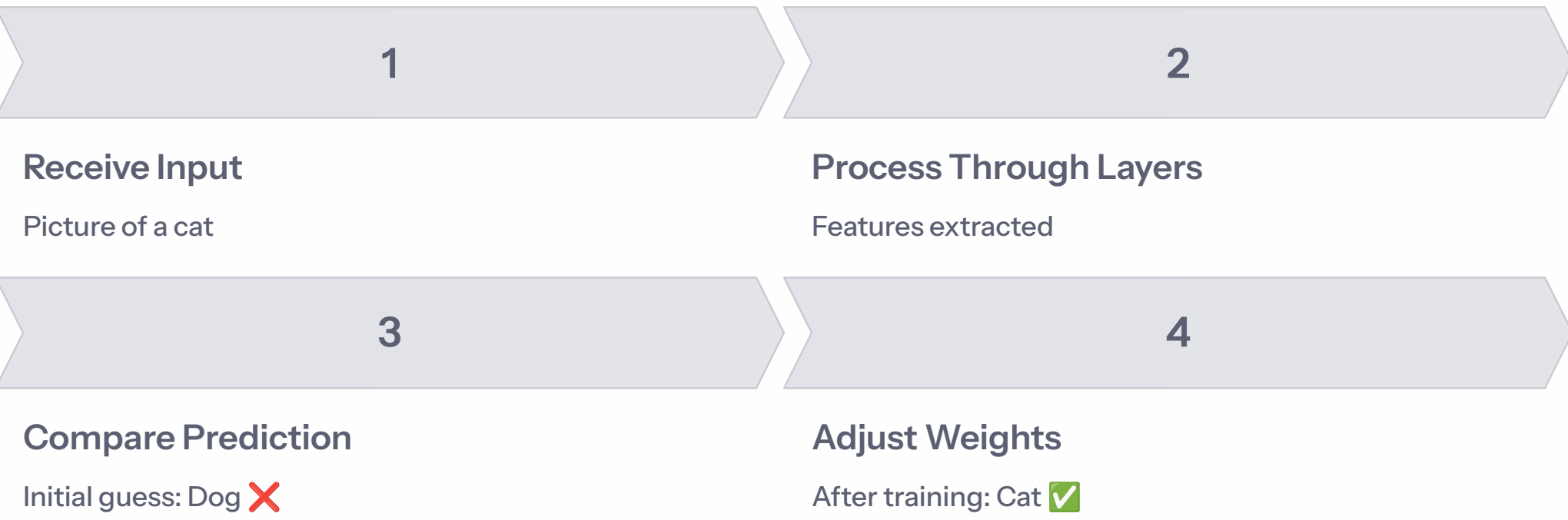


Each layer plays a distinct role — the input layer receives data, hidden layers learn complex features, and the output layer delivers the final answer.

1	2	3
Input Layer The entry point of the model. Receives raw data such as image pixels, text tokens, or audio signals.	Hidden Layers Where the actual "learning" happens. Detects edges → shapes → objects. More hidden layers = deeper learning (Deep Learning).	Output Layer Produces the final prediction — e.g., Spam/Not Spam, Cat/Dog, or a numeric house price.



3. How Neural Networks Learn



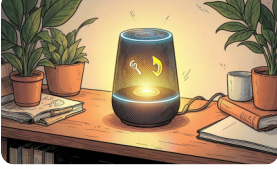
4. Deep Learning

Deep Learning is a subset of Machine Learning that uses neural networks with **multiple hidden layers**. The "Deep" in Deep Learning refers to the depth of these layers, enabling the model to learn highly complex features.



Face Recognition

Mobile unlock, security systems



Voice Assistants

Siri, Alexa, Google Assistant



Self-Driving Cars

Tesla, autonomous navigation



Chatbots

ChatGPT, conversational AI



5. CPU vs GPU

Modern AI training demands enormous computational power. Here's how CPUs and GPUs differ and why GPUs dominate deep learning.

⚙️ CPU — Central Processing Unit

- 4–8 cores (typically)
- Low latency, sequential processing
- Best for general-purpose tasks
- Web browsing, OS, scripts

One smart worker doing tasks one-by-one.

🚀 GPU — Graphics Processing Unit

- Thousands of cores
- High throughput, parallel processing
- Best for AI & Deep Learning
- Matrix calculations, image processing

Thousands of workers doing tasks together.

Feature	CPU	GPU
Core Count	Few cores	Thousands of cores
Processing Type	Sequential	Parallel
Best For	General tasks	AI & Deep Learning
Training Speed	Several days	Few hours



Final Summary

Neural Network Learns patterns from data automatically	Input Layer Receives raw input data	Hidden Layers Extracts and learns features
Output Layer Produces the final result	Deep Learning Many hidden layers = more power	CPU Sequential, general-purpose
GPU Parallel, AI-accelerated		